

TED (15) 3133

Regno

(REVISION 2015)

Signature

DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

DIGITAL COMPUTER PRINCIPLES

Model Question Paper

(Time : 3 Hrs)

(Max. Marks : 100)

PART – A

(Maximum marks : 10)

I Answer all questions. Each question carries 2 marks.

1. 2's compliment of 1011 is
2. A non-weighted codes cannot be used for arithmetic operations; State True or False
3. Define an Encoder
4. shift register can be used as Ring counter
5. List an example of PLD.

PART – B

(Maximum marks : 30)

II Answer any 5 of the following. Each question carries 6 marks.

1. Convert 1011 to gray code and gray code 1011 to Binary
2. Draw the block diagram of 4 bit Binary Parallel adder
3. Describe the applications of Multiplexers
4. Describe Race around condition in JK Flipflop
5. Design the diagram of a serial in serial out shift register using D Flip Flop
6. State the Resolution (Step size) parameter of DAC
7. Describe the types of RAMs

PART – C

(Maximum marks : 60)

(Answer one full question from each Unit. Each question carries 15 marks.)

UNIT – 1

III (a) Convert the following

	i) $(222.22)_{16}$ to Octal	
	ii) $(111110111.11011)_2$ to Hexadecimal	
	iii) 327.17 to Binary	
(b)	Expand $A + B'$ to Min terms and Max terms	6

OR

IV	(a) Describe and Prove Demorgan's Theorem	9
	(b) Describe 3 basic gates with Truth Table	6

UNIT – II

V	(a) Illustrate the working of a Serial Adder	6
	(b) Illustrate a Decimal to BCD Encoder	9
	OR	
VI	Demonstrate a 3 line to 8 line Decoder	15

UNIT – III

VII	Describe a Master Slave JK Flipflop and also demonstrate the concept to overcome the Race around Condition	15
	OR	
VIII	Demonstrate and design a 2 bit up down counter	15

UNIT – IV

IX	Describe a Successive approximation type DAC	15
	OR	
X	Describe a 4X4 RAM	15